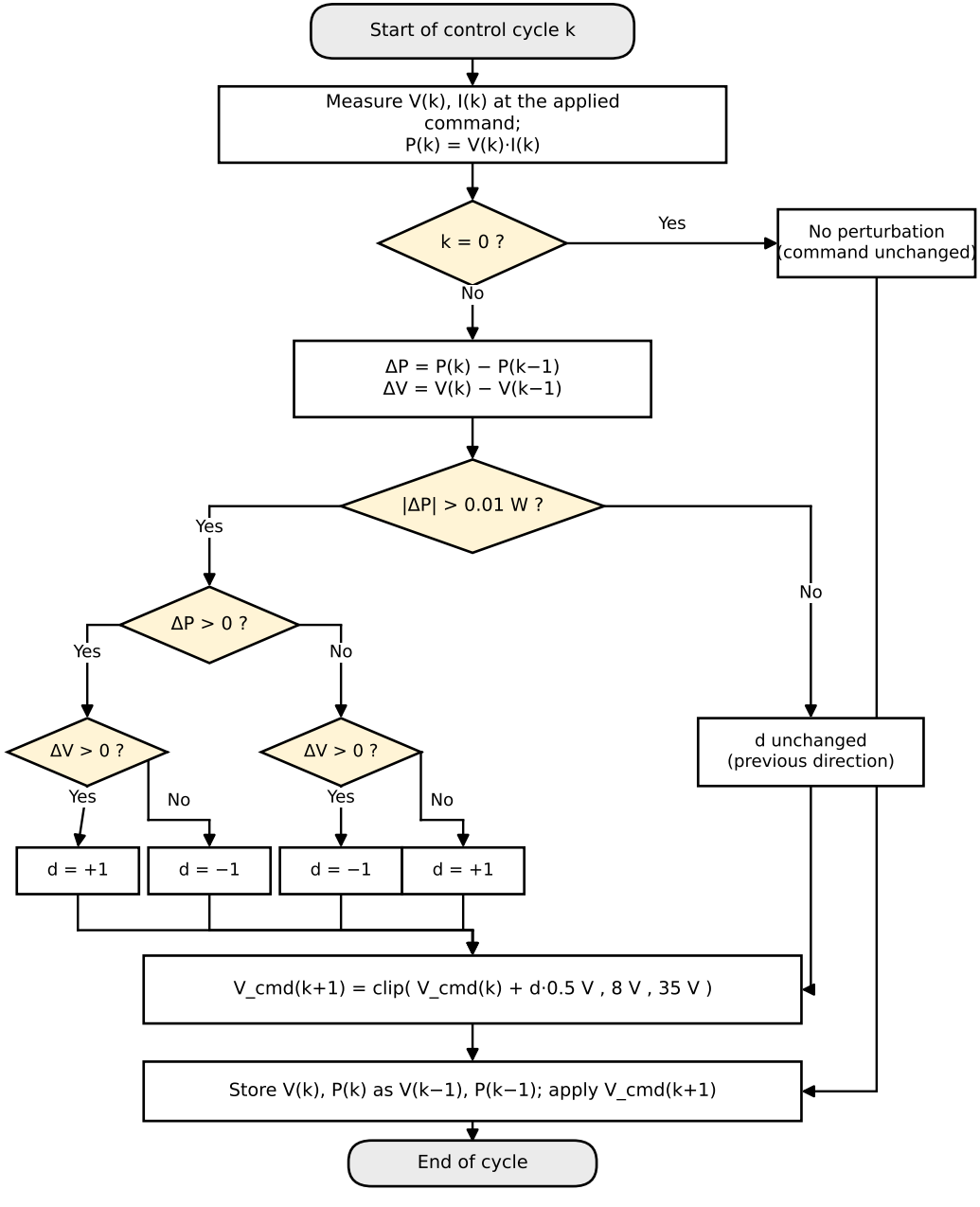
Start of control cycle k
Measure V(k), I(k) at the applied command;
P(k) = V(k)·I(k)
k = 0 ?
Yes
No perturbation (command unchanged)
No
ΔP = P(k) − P(k−1)
ΔV = V(k) − V(k−1)
|ΔP| > 0.01 W ?
Yes
No
ΔP > 0 ?
Yes
No
ΔV > 0 ?
Yes
No
ΔV > 0 ?
Yes
No
d = +1
d = −1
d = −1
d = +1
d unchanged (previous direction)
V_cmd(k+1) = clip(V_cmd(k) + d·0.5 V , 8 V , 35 V)
Store V(k), P(k) as V(k−1), P(k−1); apply V_cmd(k+1)
End of cycle